R FOR DATA ANALYSIS · R BASICS · LESSON 4

04 Comparison operators

Ask R yes or no questions with less than, greater than, equal and not equal.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Use the six comparison operators
- Read TRUE and FALSE answers
- Know the difference between == and !=

The R code in this lesson

```
# Comparison operators
# Less than: <    Greater than: >
# Less than or equal to: <=   Greater than or equal to: >=
# Is equal to: ==   Is not equal to: !=
25 < 50    # TRUE
25 > 50    # FALSE
25 <= 50   # TRUE
25 >= 50   # FALSE
25 == 50   # FALSE
25 != 50   # TRUE
```

What you will hear (times count from the end of the intro)

Intro Data analysis is full of small questions: is this bigger, is this smaller, are these the same? In this lesson, you will learn the six comparison operators in R, and how R answers each question with TRUE or FALSE. It is easier than it sounds. Let's begin.

0:00 Here I wrote a short list of notes, so we can see all six operators together. Less than, greater than, less than or equal to, greater than or equal to, is equal to, and is not equal to.

0:20 Let's try them one by one, with the same two numbers: twenty five and fifty.

0:26 Twenty five, less than, fifty. Run it. R says TRUE. Twenty five is smaller.

0:32 Twenty five, greater than, fifty. FALSE. Of course, it is not bigger.

0:40 Now less than or equal to. We write the less than sign, then the equals sign. Twenty five, less than or equal to, fifty: TRUE.

0:48 Greater than or equal to: FALSE.

0:55 Is equal to, with two equals signs: FALSE, because the numbers are different.

1:02 And is not equal to, with an exclamation mark and an equals sign. Twenty five is not equal to fifty: TRUE.

1:15 Take a moment to read the answers in the console. Every question gives TRUE or FALSE. We will use these questions a lot later, when we filter data.

1:30 Next, I add a new note: calling a function. A function is a ready-made tool in R. You give it some values, and it does the work for you.

1:50 Our first function is sort. Inside the brackets, I put a vector of numbers with c. Run it, and R lists the numbers from smallest to largest.

2:20 Then sum, which adds them all up. The answer is five thousand five hundred ninety four. We will meet more functions in the next lesson. Well done!